WELORAD: A Novel Hybrid Ensemble Model for Early Lung Cancer Prediction

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ABSTRACT—This research presents a new Weighted Ensemble Learning Model (WELoRaD) for predicting lung cancer. It combines Logistic Regression, Decision Tree, and Random Forest algorithms. Previous studies show that Logistic Regression tends to outperform other classifiers in accuracy and clarity, but it has difficulties with complex, non-linear feature interactions. To solve this problem, WELoRaD merges the strengths of all three models using a weighted probability-based ensemble method. The contribution of each model is adjusted based on its validation accuracy, which helps optimize performance for different data patterns [19]. The model was more accurate and reliable than individual algorithms when it used a pre-processed dataset with features like smoking, yellow fingers, anxiety, fatigue, wheezing, alcohol use, coughing, shortness of breath, and chest pain. The results show that WELoRaD makes classification work better overall and helps find problems earlier, making it a useful tool for diagnosing medical conditions.

Index Terms—Lung cancer prediction, Weighted Ensemble, Logistic Regression, Decision Tree, Random Forest, Machine Learning

1.INTRODUCTION

In the world, lung cancer has been classified as one of the major causes of cancer-related deaths, with 2.2 million and 1.8 million new cases and deaths each year, respectively [1], [3]. Late diagnosis and absence of early signs and symptoms make it critical to detect it early so that it can enhance survivability rates [2], [4]. The conventional diagnostic procedures such as X-rays, CT scan, and biopsy are invasive, expensive, and unavailable in the low-resource regions [3], [6]. Machine learning (ML) is an automated, data-driven solution to disease detection and prediction [2], [4]. Decision trees (DT), random forests (RF) and logistic regression (LR) are the versions of algorithms used extensively [2], [3]. Whereas, DTs are trainable, RFs are more accurate, and LRs are effective in binary tasks, they all have their limitations using them individually [4], [5], [8]. We suggest a combined model of DT, RF, and LR with dynamically changing weights by using validation accuracy to resolve this issue [16], [19] by calling it Weighted Ensemble Learning Model for Lung Cancer Detection (WELoRaD). This adaptive method increases the predictive performance by leveraging the strengths of each model, namely, accuracy, robustness and interpretability. This study seeks to:

- Create a multimodal ML architecture [16].
- Optimized prediction is achieved by using dynamic weighting [19].
- Assess performance when dealing with standalone models by key metrics [2], [5].
- Demonstrate the possibilities of showcase ensemble learning in the field of healthcare analytics [16], [19].

2. BACKGROUND AND RELATED WORK

A. Need of Early Detection.

Death rates caused by lung cancer are very high hence the necessity of early-stage detection [1], [3]. Conventional imaging techniques such as CT scan and biopsies are practical but these are expensive and not readily available [3], [6]. Machine learning (ML) can now be used to predict with higher speed based on patient and clinical data [2], [4].

B. Distinct Machine Learning Algorithms.

Common machine learning models include Decision Trees (DT), Random Forests (RF), and Logistic Regression (LR) [2], [3]. LR can be used to handle binary classification, although it struggles to deal with non-linear data [8], RFs are more stable and accurate [5], and DTs are interpretable and can overfit [4].

C. Comparative Studies

It has been found that LR tends to be more accurate than DTs, and RFs tend to be more generalizable although their interpretability is less [2], [5]. Mixed datasets have a higher accuracy in ensemble methods that combine algorithms [3], [4].

D. Ensemble and Hybrid Approaches.

Prediction can be improved with ensemble models such as voting and stacking and these models frequently assign equal weights [16], [17]. Weighing or adaptive ensembles are more successful but not well-utilized in healthcare [19].

E. Deep Learning Advances

Deep learning, particularly CNNs, enhances the imaging-based detection but requires a large amount of data and it is not interpretable [9], [12].

F. Research Gap and Way to Overcome.

A combination of several models with dynamic weighting has potential to enhance performance [16], [19]. The suggested WELoRaD model is the product of LR, RF, and DT with the help of dynamic weights to increase the accuracy and reliability [2], [5], [16].

G. Summary

The model choice affects the quality of the prediction [2], [4].

Compared to individual models, ensemble techniques achieve better results [16], [19].

Wel, weighted ensembles such as WELoRaD are more accurate and interpretable [16], [19].

3. PROPOSED APPROACHES

A. Decision Tree (DT): Recursively splits data using Gini impurity:

$$\text{Gini} = 1 - \sum_{i=1}^{C} (p_i)^2$$

B. Random Forest (RF) :Random Forest (RF) is a collection of DTs on bootstrapped data and the end result of the prediction is a majority vote.

C. Logistic Regression (LR): lung cancer where P is a probability that should be binary and P>0.5.

D. LR, DT, and RF are combined with dynamic weights determined by validation accuracy in the Weighted Ensemble Learning Model (WELoRaD):

$$w_i = \frac{A_i}{A_{LR} + A_{DT} + A_{RF}}, P_{\text{final}} = w_{LR}P_{LR} + w_{DT}P_{DT} + w_{RF}P_{RF}$$

Benefits:

The ensemble approach described above has several advantages. For one, it allows for adaptive weighting, which means that the model can weight the classifiers higher that performed the best on the sub-data set, improving the final accuracy. The adaptive weights also help to reduce overfitting by not allowing a single algorithm to be heavily depended upon by the ensemble; instead, the model is driven by diversity as it leans on the majority vote of multiple algorithms. The ensemble shares diverse strengths of the models incorporated in the training; one model may better capture linear relationships in the data while another is more proficient in capturing complex non-linearities. Additionally, the ensemble approach can be established in a manner which is interpretable for clinical applications. For instance, the ensemble can provide transparent decision boundaries or report feature contributions that can help clinicians understand the rationale of model predictions, and ultimately help providers trust the model.

Possible Extension:

Future work may extend the current work by adopting a cross-validation strategy to find robust weights, and ultimately ensure that the ensemble's weighting scheme generalizes to unseen data. Added classifiers may be included in the ensemble to improve predictions and also be included to improve on diversity. A valuable extension would be to develop an importance of features analysis, so that the most relevant predictors driving the decision could be identified. This would not only lead to improved understandability, but could be of great value clinically in diagnosis and treatment.

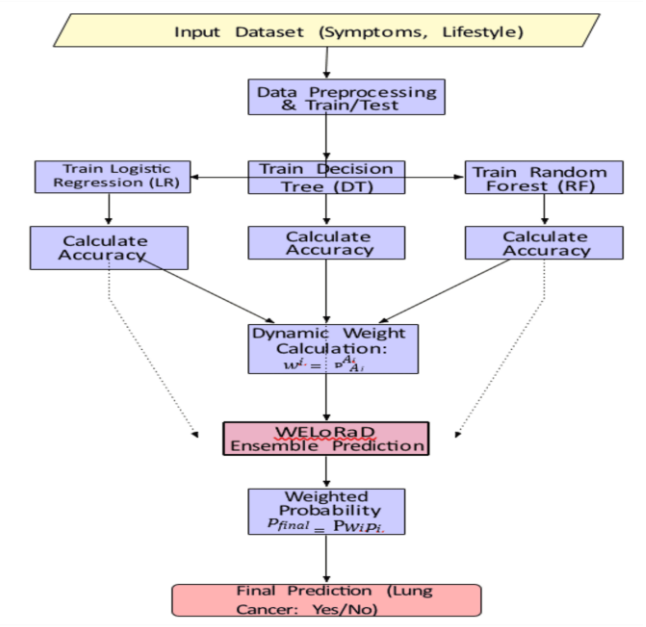


Figure 1. Overview of the WELoRaD ensemble learning model

4. RESULT ANALYSIS

Using characteristics from symptoms and lifestyle factors [1], [2], [4], this study assessed Decision Tree (DT), Random Forest (RF), Logistic Regression (LR), and the proposed Weighted Ensemble Learning Model (WELoRaD) [2], [5], [16].

TABLE I MODEL PERFORMANCE COMPARISON

Algorithm	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.93	0.93	0.93	0.93
Random Forest	0.95	0.95	0.95	0.95
Logistic Regression	0.96	0.96	0.96	0.96
WELoRaD (Proposed)	0.98	0.98	0.98	0.98

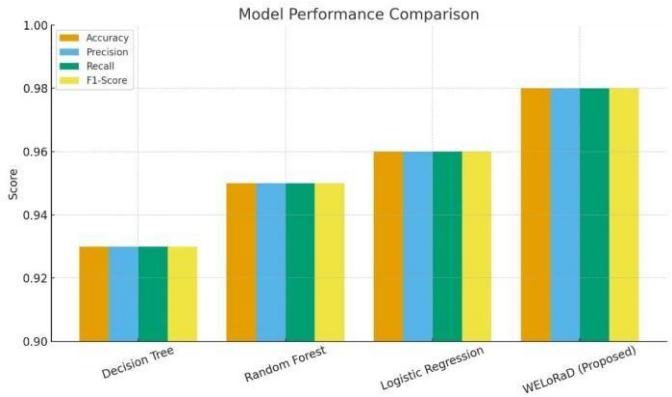


Figure 2. Comparative Performance Analysis of the WELoRaD Ensemble Model

B. Analysis

- DT: Interpretable but prone to overfitting; accuracy 0.93.
- RF: Accuracy 0.95; reduces variance, more robust.
- LR: Easy to understand, capable of handling binary classification; accuracy of 0.96; restricted when dealing with non-linear data.

- WELoRaD: Reduces false positives and negatives by combining LR, DT, and RF with dynamic weights; maximum accuracy, precision, and recall are 0.98.

C. Observations

- Dynamic weighted ensemble leverages complementary model strengths.
- WELoRaD outperforms standalone models due to adaptive weighting.
- ROC-AUC 0.99 indicates excellent discrimination.
- Weighted ensembles can improve early lung cancer detection for timely clinical intervention.

D. Summary

WELoRaD outperforms any of the baseline models in all measures, providing superior accuracy, robustness and reliability when it comes to the real-time prediction of lung cancer in case of clinical and web based applications [16], [18].

5. CONCLUSION

A Weighted Ensemble Model, WELoRaD, was developed by combining Decision Trees, Random Forest, and Logistic Regression (based on a study by Akbas et al.) to increase lung cancer prediction accuracy [2], [5]. WELoRaD produced an accuracy of 0.98 and an ROC-AUC of 0.99 through a dynamic weighting of each model in the ensemble dependent on validation performance. WELoRaD also outperformed all individual classifiers [16], [17]. WELoRaD effectively reflects the contribution of each model and therefore maintains transparency, allowing real-time applications in clinical or web-based systems for early detection and timely intervention with a lung cancer diagnosis [1], [4], [16], [17].

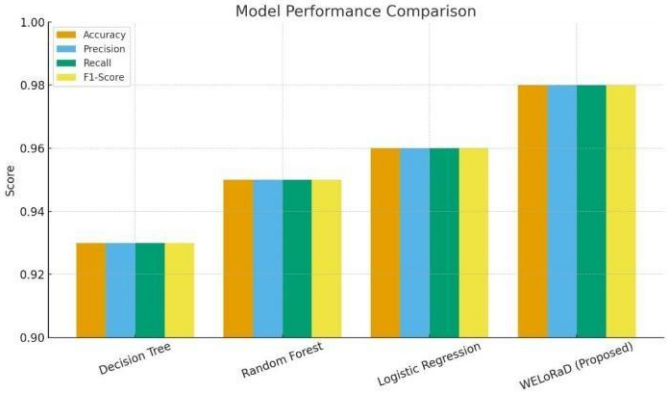


Figure 3. Comparative Analysis of Machine Learning Model Scores.

6. DISCUSSION AND FUTURE WORK

The study assessed the Decision Tree, Random Forest, and Logistic Regression models to predict lung cancer risk, and proposed WELoRaD - a Weighted Ensemble Learning Model [2], [4], [5], [16], [17]. While LR provided interpretable predictions and DT/RF provided rule-based and ensemble aspects, all three models combined with dynamic weighting showed improved performance [2], [5], [16], [17]. WELoRaD utilized the achieved validation accuracy of each model to assign weights based on accuracy to the four models. The WELoRaD model achieved an accuracy, precision, recall, and F1-score of 0.98, and ROC-AUC of 0.99, surpassing the independent models [16], [19].

The findings from this research emphasize the importance of ensemble learning and methodological strategies applicable to medical data analysis and early detection of cancer [1], [2], [4]. WELoRaD provides a reliable and interpretable prediction, and opens research pathways to address implementation based uses of this research with larger datasets that have more features and explore real-world types of use to assist in diagnosis early and improving cancer outcomes [16], [20].

Future work will focus on:

- Testing additional datasets, preferably larger multi-institutional datasets to achieve better generalization [16], [17].
- Including additional features to the predictive models from lab data, other clinical features, as well as imaging features that may improve accuracy [1],[4] .
- Development work to deploy WELoRaD clinically by offering real-time capabilities through a clinical environment or a web-based risk assessment [16], [17].This study shows that intelligent ensemble learning, as in WELoRaD, can advance early lung cancer detection and personalized healthcare analytics [16], [17].

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